Advanced JavaScript Front End Frameworks Mini Project

CIA 3 REPORT

SUBMITTED BY:

- **2462349-Prasanna RD**

  Gmail :prasanna.rd@btech.christuniversity.in
- **2462302 –Aadithya Vimal**

  Gmail : aadithya.vimal@btech.christuniversity.in
- **2462316 -Anna Isson**

  Gmail : anna.isson@btech.christuniversity.in
- 2462328-Hridyalakshmi P M

  Hridyalakshmi.pm@btech.christuniversity.in

- **COURSE**: Advanced JavaScript Front End Frameworks
- **INSTRUCTOR NAME**: *Mr .M NATH*
- **INSTITUTION**: *Christ University*
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1. ABSTRACT

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The Restaurant Ordering System is a web-based application designed to streamline restaurant operations by digitizing the process of order management, menu handling, and customer service. The system allows restaurant staff to efficiently manage table sessions, create and track orders, and monitor kitchen operations in real time. It also provides administrative features such as menu management, staff management, and analytics for business insights.

The application follows a modern full-stack architecture with a responsive user interface and secure backend services. Through role-based authentication, different users such as servers, kitchen staff, and administrators can access specific features relevant to their responsibilities. Real-time updates enable better coordination between the front-end staff and kitchen team, improving service speed and reducing manual errors**.**

The primary objective of this project is to provide an efficient digital platform that enhances restaurant workflow, improves customer experience, and simplifies management tasks through automation and data-driven insights. (github.com)

2. OBJECTIVES

The main objectives of the Restaurant Ordering System project are:

1. To develop a digital platform for managing restaurant orders and menu items efficiently.
2. To reduce manual work in restaurants by automating order processing and table management.
3. To provide real-time tracking of orders between servers and kitchen staff.
4. To implement role-based access control for administrators and staff.
5. To improve restaurant management through analytics and statistics.
6. To design a responsive user interface that works across desktops, tablets, and mobile devices.

3.TOOLS AND TECHNOLOGIES USED

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Frontend:	React, Vite, Tailwind CSS
Backend:	Java, Spring Boot
Security:	Spring Security, JWT Authentication
Database:	PostgreSQL
API Documentation	: OpenAPI (Swagger)
Build Tools:	Maven, npm
Testing:	JUnit 5, Mockito
DevOps:	Docker, Docker Compose
CI/CD:	Jenkins

4. PROJECT ARCHITECTURE & IMPLEMENTATION

The system follows a full-stack client-server architecture, where the frontend communicates with backend REST APIs. The backend handles business logic, authentication, and database operations, while the frontend provides an interactive user interface.

Architecture Layers

1. Presentation Layer ○ React-based user interface ○

   Handles user interactions and displays data

2. Application Layer ○ Spring Boot backend ○

   Processes requests and business logic

3. Data Layer ○ PostgreSQL database ○

   Stores restaurant data such as users, menu

   items, and orders

The system ensures smooth communication between these layers using REST APIs.

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4.1 COMPONENT DESIGN

The application consists of several core components**:**

1. AUTHENTICATION COMPONENT

- Handles login and registration
- Implements JWT authentication
- Ensures secure access to system resources

2. MENU MANAGEMENT COMPONENT

- Add, update, or remove menu items
- Categorize menu items
- Manage price and availability

3. ORDER MANAGEMENT COMPONENT

- Create customer orders
- Track order status
- Send order information to the kitchen

4. STAFF MANAGEMENT COMPONENT

- Manage staff accounts
- Assign roles and permissions

5.ANALYTICS COMPONENT

- Display revenue statistics
- Track popular menu items
- Generate operational insights

4.2 ROUTING STRATEGY

Configured Angular Router for seamless navigation:

The routing mechanism enables smooth navigation between different pages of the application without reloading the entire interface. It ensures that users can easily move between different

system modules such as the dashboard, menu management page, order management page, and analytics section.

Routes are protected using authentication guards to ensure that only authorized users can access restricted features such as administrative dashboards**.**

Key routing modules include:

- Dashboard view
- Menu management page
- Order management page
- Kitchen queue dashboard
- Staff management panel

4.3 FORMS AND VALIDATION

Forms play an important role in collecting and validating user input in the system.

Examples include:

- User login and registration forms
- Menu item creation forms
- Staff registration forms
- Order placement forms

Validation techniques include:

- Required field validation
- Email format validation
- Password security checks
- Numeric validation for pricing and quantities

These validations ensure data accuracy and prevent invalid data from being stored in the database.

5. KEY FEATURES

The Restaurant Ordering System provides several important features:

1. **USER AUTHENTICATION**
   - Secure login and registration using JWT**.**
2. **ROLE-BASED ACCESS CONTROL**
   - Different access privileges for admins and staff.
3. **MENU MANAGEMENT**
   - Add, update, and remove food items.
4. **ORDER MANAGEMENT**
   - Create and manage customer orders.
5. **KITCHEN QUEUE DASHBOARD**
   - Real-time tracking of pending orders.
6. **TABLE SESSION MANAGEMENT**
   - Track customer sessions and billing.
7. **ANALYTICS DASHBOARD**
   - View revenue and operational statistics.
8. **RESPONSIVE DESIGN**
   - Works on desktops, tablets, and mobile devices.

These features help automate restaurant workflows and improve efficiency. (GitHub)

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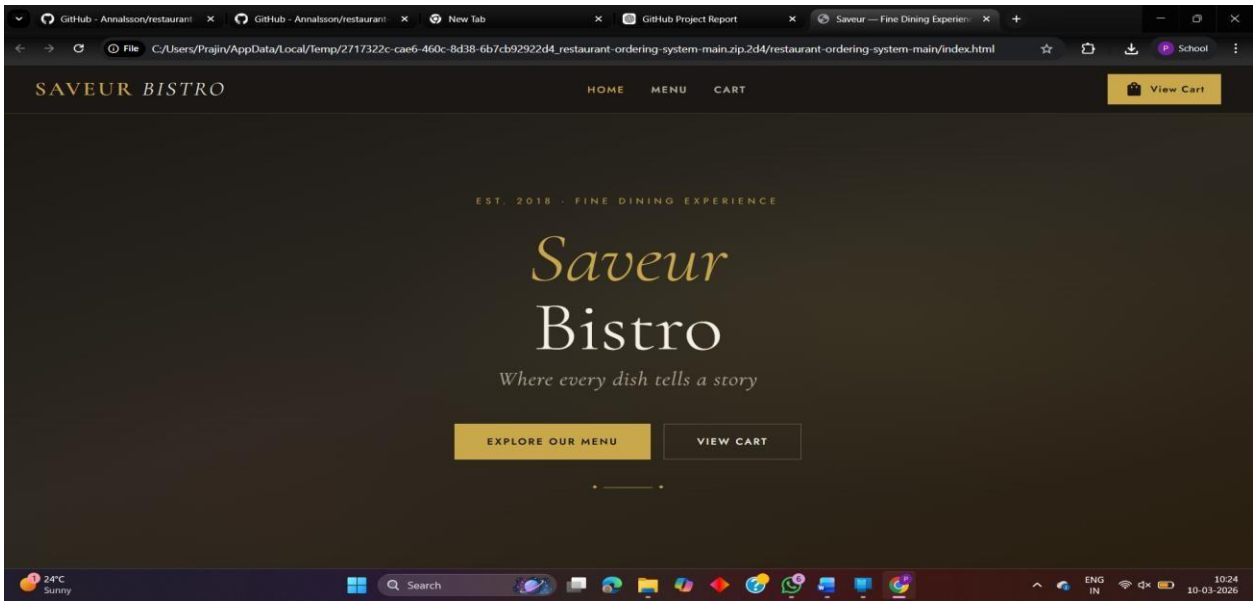
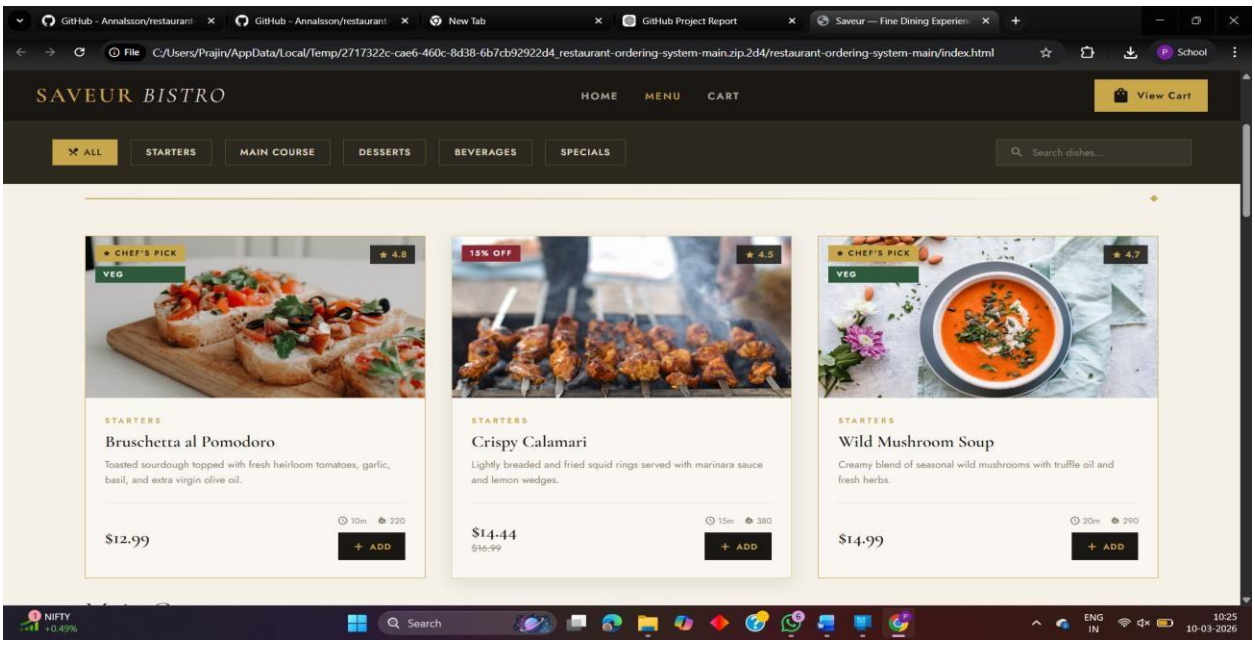
7.SCREENSHOTS OF FINAL OUTPUT



6.1ARTIST DISCOVERY VIEW

This section displays a dashboard that allows users to explore available items or system modules. It provides a structured interface where users can easily navigate through different restaurant management functions.

SCREEN SHOT:



6.3 GLOBAL NAVIGATION

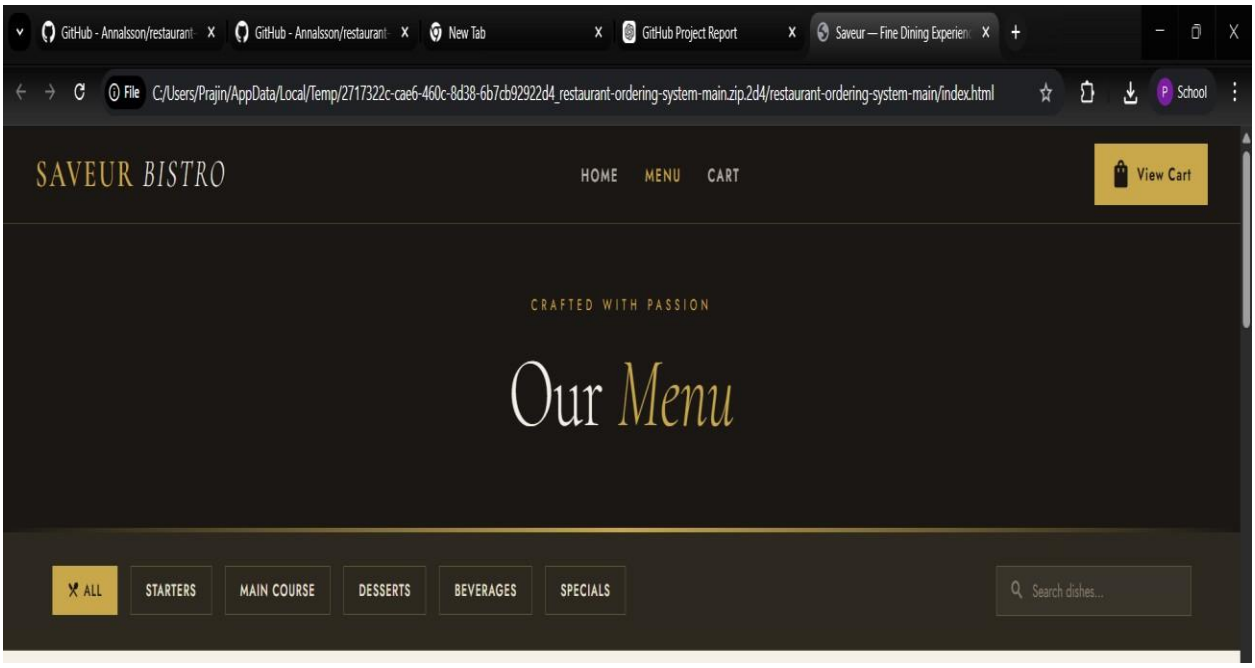
The global navigation system provides a menu or sidebar that allows users to quickly move between different sections of the application such as:

- Dashboard
- Menu Management
- Orders
- Staff Management
- Analytics

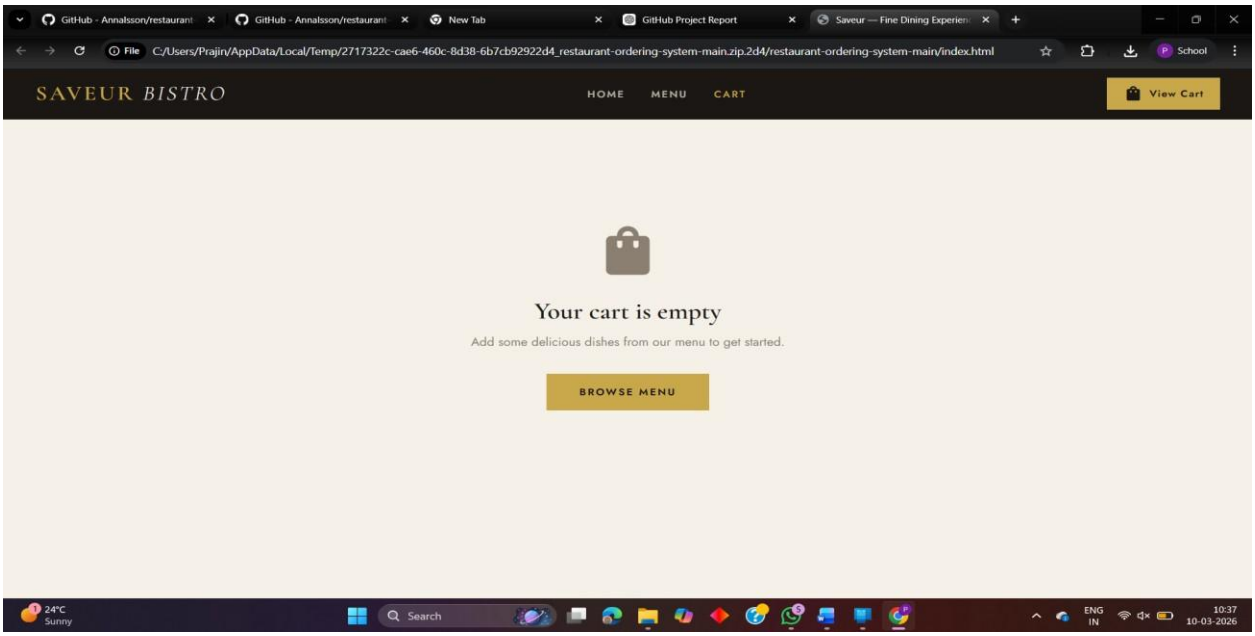
This ensures a smooth and user-friendly navigation experience.

SCREEN SHOT:

1.Dynamic menu :



2.Your Cart :



7. SOURCE LINK : GitHub

Repository:

https://github.com/AnnaIsson/restaurant-ordering-system.git

8.CONCLUSION

The Restaurant Ordering System provides a comprehensive solution for managing restaurant operations digitally. By integrating modern technologies such as Spring Boot, React, and PostgreSQL, the system offers a secure, scalable, and efficient platform for handling restaurant workflows.

The system improves communication between servers, kitchen staff, and administrators by enabling real-time order tracking and automated management processes. Additionally, analytics features help restaurant owners gain valuable insights into their business performance.

Overall, the project demonstrates how modern web technologies can be used to develop a reliable and efficient restaurant management system that enhances operational efficiency and customer satisfaction.

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